

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Summer Examination – 2025

Course: B. Tech

Branch: Civil Engineering

Semester: VII

Subject Code & Name: BTCVC704_Y20 & Professional Practices

Max Marks: 60

Date: 24/07/2025

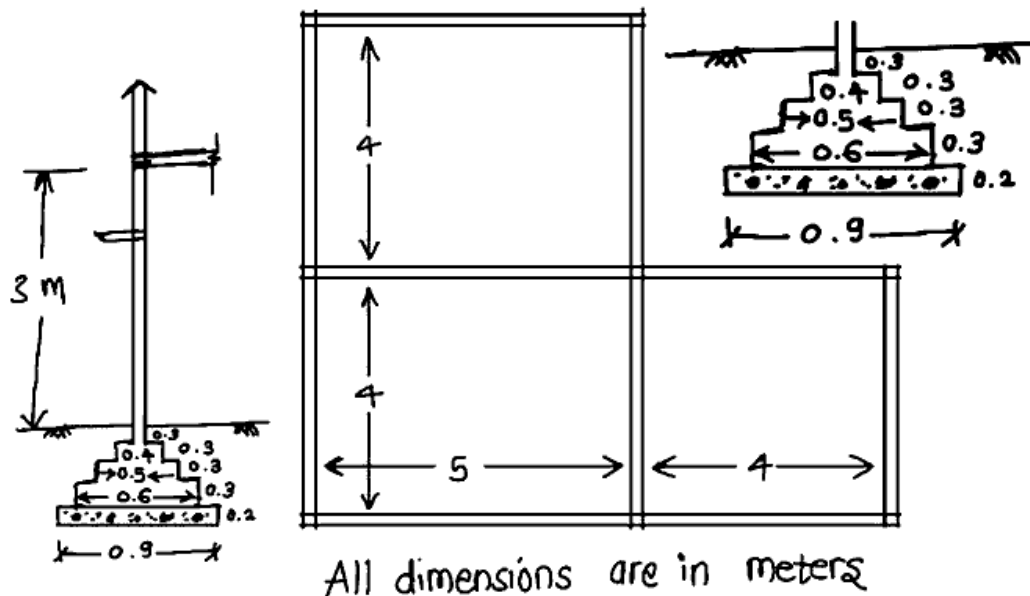
Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level /CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 Which of the following is NOT included in a detailed estimate?		1
A) Quantity of work	B) Rates of items	
C) Equipment specifications	D) Cost of each item	
2 The purpose of technical sanction is to:		1
A) Approve the location of the site		
B) Finalize material suppliers		
C) Ensure the technical soundness of the estimate		
D) Approve work completion reports		
3 Which method is commonly used for taking out quantities in P.W.D. practice?		1
A) Centre line method	B) Crossing method	
C) Midpoint method	D) Diagonal method	
4 Price escalation in construction is mainly due to:		1
A) Use of high-grade materials only	B) Fixed labor costs	
C) Fluctuation in market prices	D) Reduced project duration	
5 Which type of estimate is most accurate and used for preparing tender documents?		1
A) Plinth area estimate	B) Cube rate estimate	
C) Preliminary estimate	D) Detailed estimate	

- 6 Standard Schedule of Rates (SSR) is mainly used to: 1
- A) Design structures B) Define working hours
- C) Fix payment schedules D) Determine unit rates of construction items
- 7 Which of the following is a type of execution model in infrastructure projects? 1
- A) BOQ B) BOT
- C) SSR D) RCC
- 8 The purpose of competitive bidding is to: 1
- A) Increase cost B) Avoid contract
- C) Reduce contractor accountability D) Ensure fair selection of contractor
- 9 In P.W.D. system, which document is used for recording payments to contractors? 1
- A) Rate list B) Tender notice
- C) Measurement book D) Bill form only
- 10 Arbitration is a method used for: 1
- A) Site selection B) Conflict resolution outside court
- C) Price fixation D) Quantity estimation
- 11 Which method of valuation is based on the net income from the property? 1
- A) Cost method B) Market comparison method
- C) Income capitalization method D) Development method
- 12 Obsolescence refers to: 1
- A) Loss due to natural calamity
- B) Appreciation of asset value
- C) Loss in value due to outdated design or function
- D) Tax applied to properties
- Q.2 Solve the following. 12**
- A) Define estimating. What is its importance in civil engineering projects? 6
- B) Calculate the following quantities from figure. Take wall thickness as 0.3 meters 1. 6
- Earthwork in excavation 2. PCC in foundation, 3. Brickwork in foundation and Superstructure (Do not consider the deductions)



- Q.3 Solve the following.** **12**
- A) Prepare abstract sheet for data calculated in Q 2B and carry out rate analysis. Assume appropriate/suitable rates per unit. 6
- B) What is price escalation? How is it accounted for in estimating and costing of civil works? 6
- Q.4 Solve Any Two of the following.** **12**
- A) What are the major components of a typical tender paper? Discuss in brief. 6
- B) What is the role of competitive bidding in the tendering process? 6
- C) Write short notes on B.O.T. projects with suitable examples from infrastructure works. 6
- Q.5 Solve Any Two of the following.** **12**
- A) Explain the organizational setup of the Public Works Department (P.W.D.). 6
- B) What is arbitration? When is it used in construction contracts? 6
- C) State the essential elements of a legally valid contract. 6
- Q.6 Solve Any Two of the following.** **12**
- A) What is the difference between value, price, and cost in the context of property valuation? 6
- B) What is a sinking fund? How is it used in property valuation? 6
- C) Describe any two methods of calculating depreciation. 6

*** End ***